INCH-POUND
MIL-D-24542B
15 September 1988
SUPERSEDING
MIL-D-24542A
31 January 1980
(See 6.9)

MILITARY SPECIFICATION

DIVING OUTFIT, LIGHTWEIGHT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a lightweight diving outfit utilizing a demand breathing device for use in diving operations, employing two divers and a standby diver to 190 feet when air is used for breathing and, additionally) when supported by an open diving bell to 300 feet when helium-oxygen gas mixture is used for breathing.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FEDERAL (Continued)

PPP-B-665 - Boxes, Paperboard, Metal Edged and Components.
PPP-B-676 - Boxes, Setup.
PPP-C-1842 - Cushioning Material, Plastic, Open Cell (For Packaging Application).
PPP-P-40 - Presentation and Packing of Hand Tools; Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-117 - Bags, Sleeves and Tubing - Interior Packaging.
MIL-R-196 - Repair Parts, Accessories, and Kits, Mechanical; Packaging of.
MIL-B-233 - Boxes, Supply Support Items, Stowage and Storage.
MIL-P-775 - Packaging of Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing); and Fittings, Nozzles, and Strainers.
MIL-C-3131 - Cordage; Packaging of.
MIL-C-12000 - Cable, Cord, and Wire, Electric; Packaging of.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of.
MIL-B-81705 - Barrier Materials, Flexible, Electrostatic-Free, Heat Sealable.
MIL-P-81997 - Pouches, Cushioned, Flexible, Electrostatic-Free Reclosable, Transparent.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MS90363 - Box, Fiberboard with Cushioning for Special Minimum Cube Storage and Limited Reuse Application.

2.1,2 Other Government drawings and Publications. The following other Government drawings and publication form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the. date of the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

NAVSHIPS DL 4666881 - Divers Mask USN MKl MOD O.
NAVSHIPS DL 4666883 - Umbilical, Divers Mask USN MKl MOD O.
NAVSHIPS DL 4666884 - Setup/Maintenance Kit, Divers Mask USN MKl MOD O.
NAVSHIPS DL 4666885 - Divers Intercom.
NAVSHIPS DL 4666887 - Operating Specifications, Divers Mask USN MKl MOD O.

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NAVAL SEA SYSTEMS COMMAND (NAVSEA) (Continued)

NAVSHIPS DL 4666950 - Umbilical, Divers Mask M45K USN MKl MOD O 300 Ft, 3/8".

NAVSHIPS DL 4666989 - Assembly, Integrated Diver's Vest.

PUBLICATION

N A V S E A

OD 46363 - Requirements for Electrostatic Discharge Protection of Electronic Components and Assemblies.

(Application for copies should be addressed to Code 032, Naval Ocean Systems Center, San Diego, CA 92152,)

(Copies of specifications, standards, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation, Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation,

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept. , 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of Precedence, In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Construction. The lightweight diving outfit shall consist of equipment necessary to support a diving operation employing two working divers and one standby diver. The outfit shall include four diver's mask assemblies (see 3.3), three diver's umbilical assemblies (see 3.4), three integrated diver's vests (see 3.5), a diver's intercommunications set (see 3.6), a setup and maintenance kit (see 3.7), and operating space items (see 3.7). A reusable container shall be furnished that provides stowage for certain parts of the outfit (see 5.1.2.1).

3.3 Diver's mask assembly. The diver's mask assembly shall be in accordance with the drawings listed on DL 4666881.

3.3.1 Special tooling. If specified (see 6.2.1), molds, jigs, and dies for production of the mask frame and of the oral-nasal mask will be made available to the contractor "as is, and where is" by the Government. This special tooling will be available for viewing (as is, where is) by prospective bidders on a contract for the purpose of preparation of their bids. At completion of a contract under which the contractor has taken delivery of the special tooling, the special tooling shall be redelivered to the Government in condition "as good as received, less normal wear and tear".

3.3.2 Mask control. The mask control shall withstand mask control performance as specified (see 4.6.2).

3.4 Diver's umbilical assembly. The diver's umbilical assembly shall be in accordance with the drawings listed on DL 4666883 and DL 4666950. The assembly shall withstand continuity as specified (see 4.6.1). When specified (see 6.2.1), this item shall be delivered disassembled.

3.5 Integrated diver's vest. The integrated diver's vest shall be in accordance with the drawings listed on DL 4666989.

3.6 Diver's intercommunications set. The diver's intercommunications set will be an item furnished by the Government (GFE, see. 3.10). The essential mechanical and dimensional characteristics of this item are shown on Drawing 4666885.

3.7 Setup and maintenance kit. The setup (including tools) and maintenance kit shall be in accordance with the drawing listed on DL 4666884.

3.8 Operating space items. Operating space items shall include those items listed on DL 4666887, in the quantity indicated thereon, the demand regulator operating space item kits (see 3.8.1 and 3.8.2), and the cylinder valve operating space item kit (see 3,8.3).

3.8.1 Demand regulator operating space item kit. The demand regulator operating space item kit shall contain twelve replacements for each part of the demand regulator that is fabricated entirely or in part from plastic or rubber, with the exception of the exhaust tube and the handwheel. The kit shall contain

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two replacement handwheels, two replacement exhaust tubes, four replacement springs for each spring used in the demand regulator, and four replacement actuating levers (horseshoes). The kit shall also contain six replacement inlet fittings (low-pressure nozzles or inlet nipples) and six replacements for each nut and each spring type of retainer used in the demand regulator. The individual parts, unit packaged in accordance with the requirements of 5.5, shall be additionally marked with their identifying numbers as used on the demand regulator maintenance instruction sheet (see 3.11), and consolidated with the maintenance instruction sheet into kit form. This kit shall be marked with a manufacturer's part number.

3.8.2 First stage regulator operating space item kit. The first stage regulator operating space item kit shall contain twelve replacement parts for each replaceable part of the first stage regulator that is fabricated entirely or in part from plastic or rubber, with the exception of the flexible hose assembly and securing knob, The kit shall contain four replacement flexible hose assemblies (that is, hose with end fittings), four replacement high-pressure nozzles (or that smallest or most basic subassembly that contains the high-pressure nozzle), four replacement spring type of retainers, four replacement regulator inlet (sintered) filters, four replacement relief valves (or that smallest or most basic subassembly that contains the relief valve) , and two replacement knobs. The individual parts, unit packaged in accordance with the requirements of 5,5, shall be additionally marked with their identifying numbers as used on the first stage regulator maintenance instruction sheet (see 3.11) and consolidated with the maintenance instruction sheet into kit form. This kit shall be marked with a manufacturer's part number.

3.8.3 Cylinder valve operating space item kit. The cylinder valve operating space item kit shall contain twelve replacement parts for each replaceable part of the valve assembly that is fabricated either entirely or in part from plastic or rubber and twelve replacements for each washer, with the exception of the on and off handwheel; there shall be two replacement handwheels. There shall also be four replacement valve stems and retainer nuts (or other retainer that holds the valve handle to the valve stem) and four of each piece required to completely replace the safety valve (or safety plug). The individual parts, unit packaged in accordance with the requirements of 5,5, shall be additionally marked with their identifying numbers as used on the cylinder valve maintenance instruction sheet (see 3.11), and consolidated with the maintenance instruction sheet into kit form, This kit shall be marked with a manufacturer's part number.

3.9 Controlled source part. On all equipment delivered under one contract, a part that is described by source control documentation shall be of a single source and model.

3.10 Government furnished. The following items are Government-furnished properties for incorporation into the lightweight diving outfit furnished under this specification,

(a) Diver's intercommunications set (see 3.6).
(b) Diver's intercommunications set operations and technical manual (see 3.12).
(c) Lightweight diving outfit technical manual (see 3.12).

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The contractor shall notify the contracting activity of his requirement for each of items (a), (b), and (c), and of the last acceptable date for delivery of these items to his plant for incorporation into the lightweight diving outfit. Such notification shall be given in time to allow 180 days for delivery of the items to the contractor's plant by the Government before the last acceptable date.

3.11 Regulators and cylinder valve maintenance instructions. A copy of each of the manufacturer's commercial maintenance instructions, including illustrated parts breakdown (exploded view), for the demand regulator, for the first stage regulator and for the cylinder valve used in the outfit shall be furnished packaged with the operating space item kits for these items (see 3.8.1, 3.8.2, and 3.8.3).

3.12 Technical manuals. Two technical manuals shall be packaged with the diving outfit, one covering the lightweight diving outfit in general, and the other covering the diver's intercommunications set. These manuals will be items furnished to the contractor by the Government (GFP, see 3.10). They shall be packaged with the outfit.

3.13 Workmanship. Equipment shall be free of defects such as cracks or flaws in material. Made-up joints shall be leak-tight. Air (gas) channels shall be clear and free of collars (or steps), burrs, and other protrusions into flow paths . There shall be no gross tool marks. Mold flashings shall be removed. Polished external surfaces shall have a consistency and evenness of finish. Assembled equipment shall be free of chips, filings, and other residues resulting from manufacturing processes, Internal components, parts, and pipes shall be cleaned of grease, oil, and other material deleterious to its use with high pressure air. Except where otherwise specified, colors shall be even. The equipment shall be free of burrs, sharp corners, and edges that will injure human skin or puncture pure gum sheet rubber.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are 'deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract, Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 Inspection system. When specified in the contract or order, an inspection system program plan shall be prepared (see 6.2.2) .

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be performed on one complete lightweight diving outfit and shall consist of the examination and tests specified in table I.

TABLE I. First article inspection.

Examination and test	Requirement	Test
Visual examination	3.13	4.5
Umbilical continuity	3.4	4.6.1
Mask control performance	3.3.2	4.6,2
Packaging inspection	-----	4,7

4.3.1 First article inspection report. When specified in the contract or order, a first article inspection report shall be prepared (see 6.2.2).

4.4 Quality conformance inspection, Quality conformance inspection shall consist of the production inspection of 4.4.1.

4.4.1 Production inspection. Each diving outfit shall be visually examined in accordance with 4.5 and each mask shall be subjected to the mask performance test of 4.6.2.

4.5 Visual examination. The sample shall be visually examined to assure conformance with the requirements of 3.13.

4.6 Tests.

4.6.1 Umbilical continuity. Electrical continuity between corresponding pins on opposite ends of the communications cable shall be determined.

4.6.2 Mask control performance. The controls of each sample mask (see figure 1) shall be subjected to the following test, using dry water-pump air or nitrogen in the sequence specified:

(a) Place the mask on a test fixture that will isolate the oral-nasal cavity from the mask frame cavity and both cavities from ambient pressure. Attach a compressed air supply with a capacity of at least 40 standard cubic feet per minute (ft^3/min) in the gauge pressure range of 60 to 180 pounds per square inch (lb/in), being adjustable for pressure within that range, to the breathing air inlet of the mask (see figure 1).

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(b) With valves A and B closed, set the air supply to a gauge pressure of 60 plus O, minus 10 lb/in^2, and set the demand regulator adjustment to the full-out position. The regulator shall be free flow.

(c) Reset the demand regulator adjustment to the full-in position and the air supply to a gauge pressure of 180 plus 10, minus O lb/in^2. The regulator shall not free flow.

(d) Reset the air supply to a gauge pressure of 125 ± 5 lb/in^2 and open valve B all the way. Flow into the mask through valve B shall be not less than 35 ft^3/min.

(e) Close valve B. Reset the demand regulator adjustment so that the regulator starts to deliver air when a partial vacuum of 1.50 ± 0.10 inches of water is applied to the oral-nasal cavity. Increase the partial vacuum in the oral nasal cavity to 300 ± 0.10 inches of water. The flow into the mask through the regulator shall be measured to be not less than 15 ft^3/min.

(f) Reset the air supply to a gauge pressure of 180 plus 10, minus O lb/in^2. Adjust the demand regulator adjustment so that the regulator does not free flow. Immerse the mask and test fixture in water to cover. Depress the regulator purge button until there is a steady airflow from both the mask frame exhaust valve and the regulator exhaust valve and the regulator exhaust tube, The mask shall be free of air leaks (air leaks that can be attributed to the text fixture or to its seals with the mask are of no consequence).

4,7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein,

4.7.1 First article sample. The contractor shall conduct inspection on one complete package, packed for shipment, to ascertain that the cleaning, drying, preservation, packaging, packing, and marking of the equipment or item conforms to this specification and the quality conformance inspection requirements of MIL-B-233. First article inspection shall be repeated when changes in preservation and packing materials, or processes, or designs are made.

4.7.2 Dummy load. When specified (see 6.2.1), a dummy or simulated load, packaged and packed for shipment, may be used for the first article inspection (see 4.7,1).

4.8 Government examination of last five items. In the event that Government-furnished special tooling (see 3.3,1 and 6,2.1) has been used in production of equipment, the last five mask frames and oral-nasal masks produced will be examined by the Government inspector to determine that the condition of Government-furnished special tooling is in accordance with the requirements of 3.3.1.

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.7.)

5.1 General requirements.

5.1.1 Drawingd. When specified in the contract or order, complete description details on drawings, preservation, packing, and marking shall be prepared (see 6.2.2).

5.1.2 Container design criteria. Lightweight diving outfits shall be packaged in a three-container concept as specified herein.

5.1.2.1 Container number one. The container shall conform to type M of MIL-B-233, except for size and the following:

(a) The container size shall be the minimum necessary to pack the following equipment of the outfit:

(1) Four diver mask assemblies without backplate and cylinders consisting of mask with hood, spider, and insert (see table II, note 2).
(2) Three integrated diver's vest.
(3) One diver's intercommunications set,
(4) One setup, maintenance kit.
(5) One kit of miscellaneous parts for the umbilical assembly (such as D-rings, tape, and tie cord).
(6) Operating space items (see 3.8 and 5.5).

(b) The container shall be provided with accessory framing trays, partitions, compartments , or other means that will provide space for parts storage with ease of removal or replacement.
(c) The container shall be provided with ventilation, splash protection, and drainage with all openings covered with a corrosion-resistant wire cloth or screening that will prevent entrance of vermin or rodents.
(d) Nominal 2- by 4-inch wood skids shall be bolted to the base of the container.
(e) Rope handles shall not be used.

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TABLE II. Preservation (methods or sub-methods).

Item subassembly	Method or sub-method of MIL-P-116 1/	Other packaging process specification
Diver's mask	IC-3 2/	-----
First stage regulator and hose	IC-3	-----
Safety harness	III	-----
Setup and maintenance kit	-----	PPP-P-40
Intercom box	IC-3	-----
Cylinder with filler valve	III 3/	-----
Cylinder harness	III	-----
Divers umbilical		
Assembled		
Disassembled:	III	-----
Marrying tape	IC-1	-----
Telecord splice kit	IC-3	-----
Telecord	-----	MIL-C-12000
Lifeline with finished ends, D-rings and belt hook	-----	MIL-C-3131
Hose, air or pneufathometer with fittings attached	-----	MIL-P-775
Marrying cord	III	-----
Integrated diver's vests	III	-----

1/ Bags, sleeves, or envelopes utilized for methods IC-1, IC-3, and method III shall be of transparent material as specified in MIL-P-116.
2/ Each diver's mask and hood assembly, before packaging (see 5.2.1.2 and 5.2.2), shall be furnished with an insert to provide support to prevent creasing or crushing of the hood material, and to serve as a handling, drying and storage fixture.
3/ Cylinders shall be pressurized a minimum gauge pressure of 3 lb/in^2 with dry air.

5.1.2.2 Container number two. The container shall conform to 5.3 for the level specified and shall contain four cylinders with filler valves and four cylinder harnesses. Cylinders shall be assembled to the harness.

5.1.2.3 Container number three. The container shall conform to 5.3 for the level specified and shall contain three complete divers umbilicals, assembled or disassembled as required (see 3.4 and 6.2.1).

5.1.3 Closure of openings. All openings into valves, valve breathing mechanisms, or hose ends (with or without fittings), shall be sealed with plastic caps or by wrapping the opening with a flexible, waterproof barrier. The flexible, waterproof barrier shall be secured with waterproof pressure-sensitive tape to prevent entrance of dust, dirt, or any foreign material.

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5.1.4 Cushioning, anchoring, blocking, and bracing. Cushioning, anchoring, blocking, and bracing of the contents of all containers shall be provided to ensure that the contents are protected against damage from abrasion or deformation during handling, shipment, and storage. Guidelines for cushioning, anchoring, blocking, and bracing are specified in MIL-STD-1186. Excelsior, shredded newspaper, and similar materials that have not been chemically refined for noncorrosiveness are unacceptable unless they are enclosed in a sealed waterproof barrier, bag, or envelope. Loose-fill material is prohibited for packaging and packing applications such as cushioning, filler, or as dunnage material.

5.1.5 Use of dusting material. Talc or talcum and soapstone powders used in the packaging process of items shall be free of asbestos.

5.2 Preservation. Presentation shall be level A or C, as specified (see 6.2.1).

5.2.1 Level A.

5.2.1.1 Cleaning and drying. Equipment of the diving outfit shall be thoroughly cleaned and dried in accordance with the applicable requirements of MIL-P-116 which will ensure that all parts are clean, dry, and free of grease, oil, and any other foreign matter that might adversely affect the operation of the outfit.

5.2.1.2 Unit protection. Unless otherwise specified (see 6.2.1), assemblies of each lightweight diver's outfit shall be individually unit protected in accordance with the requirements of table IX.

5.2.1.3 Unit and intermediate containers. Like items, unit-protected as specified in table II, may be individually packaged in unit or combined as intermediate containers. Unit and intermediate containers shall be of folding, set-up paperboard, fiberboard, or metal-stayed paperboard conforming to PPP-B-566, PPP-B-636, PPP-B-665, and PPP-B-676, respectively, with selection at the option of the contractor, The gross weight of paperboard boxes shall not exceed 10 pounds, Unless otherwise specified (see 6.2.1), the gross weight of fiberboard boxes shall not exceed 20 pounds. Box closure shall be as specified in the applicable box specification or appendix thereto, except when class weather-resistant boxes conforming to PPP-B-636 are used, method V closure in the appendix to PPP-B-636 shall apply.

5.2.2 Level C. Packaging of lightweight diving outfits shall afford protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity for immediate use. The contractor's normal preservation-packaging q ethods may be utilized when such meet the requirements of this level. In addition, the diver's mask and hood shall, as a minimum, be provided with an insert as specified under table II (note 2).

5.2,3 Government-furnished special tooling. If the contractor has taken delivery of Government-furnished special tooling (see 3.3.1), at the completion of the contract, he shall reclean the special tooling, represerve it in its original manner, and return it to its packaging, packing, and marking. The packaging, packing and marking shall be restored to the "as new" condition.

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5.3 Packing. Packing shall be level A, B, or C as specified (see 6.2.1).

5.3.1 Levels A and B. Exterior (shipping and storage) containers shall conform to table III. Unless otherwise specified (see 6.2.1), the container selection under the level of packing specified shall be at the contractor's option. Container number one (see 5.1.2.1) shall not require overpacking.

TABLE III. Container selection for Packing. 1/ 2/

Specification	Container	Packing application	
		Level A	Level B
		(Style, type, or class)	
PPP-B-601	Wood, cleated plywood	Overseas type	Domestic type
PPP-B-621	Wood, nailed and lockcorner	Class 2 - overseas	Class 1 - domestic

1/ Maximum gross weight, container plus contents, shall not exceed the applicable requirements for the style, type, or class container selected for the level of packing to be applied.
2/ Shipping containers shall be closed, strapped, or banded in accordance with the applicable container specification, or appendix thereto.

5.3,2 Level C. Lightweight diving outfits packaged as specified shall be packed in containers acceptable to the common carrier that will ensure safe delivery at the destination in a satisfactory condition at the lowest applicable rate. Containers or method of packing shall conform to Uniform Freight or National Motor Freight classification rules or regulations or other carrier rules as applicable to the mode of transportation.

5,4 Marking. In addition to any special marking required (see 6.2.1), marking shall be in accordance with MIL-STD-129 and MIL-P-116.

5.4.1 Special marking. In addition to the marking required by MIL-B-233, container number one (see 5.1.2.1) shall be marked "REUSABLE CONTAINER - DO NOT DESTROY".

5.5 Operating space items and tools. Operating space items and tools shall be preserved, packed, and marked in accordance with the levels of PPP-P-40, MIL-C-3131, MIL-R-196, or MIL-E-17555, as applicable, for the intended use and destination as shown in table IV.

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TABLE IV. Presevation and Packing of operating space items and tools.

Intended use - destination	Levels	
	Preservation	Packing
Accompanying equipment	A	1/
Stock	A	B
Immediate use	C	C

1/ Unless otherwise specified (see 6.2.1), operating space items shall be packed in container number one (see 5.1.2.1).

5.5.1 Repairable, Repairable operating space items subject for return to a repair facility for restoration and reissue shall be packaged and packed in materials and containers able to be reused. Unless otherwise specified (see 6.2.1), packaging shall be in accordance with MS90363. Reusable containers, when required repairable, shall be in accordance with the guidelines of MIL-E-17555. Containers shall be marked "REUSABLE - DO NOT DESTROY". When specified in the contract or order, special packaging instructions shall be prepared (see 6.2.2) ,

5.6 Sensitive electronic item protection.

5.6.1 Sensitive electronic item protection. Items such as, but not limited to, diodes, transistors , integrated circuits, and equipments incorporating such items that are susceptible to damage from electrostatic, electromagnetic or both field forces shall be protected against such damage as follows:

5.6.2 General requirements.

(a) Lead and terminal protection. Leads or terminals configuration shall be maintained as manufactured without causing loads or stresses able to cause damage to the item. Protection shall be by means of a carrier, container configuration, or inserts of noncorrosive electrostatic free supporting materials. Materials used to maintain item position and lead or terminal configuration shall permit item removal and replacement without damage to the item,

(1) Carrier. When used for additional protection of miniature electronic items, carriers shall be of such strength to prevent damaging resonances, shocks, and electrostatic charges to the sensitive items. Anchoring or securing of the item, leads or terminal within the carrier by means of tape or adhesive is prohibited. The carrier shall maintain physical separation and manufactured configuration of the item leads or terminals during packaging, handling, transportation, storage and stowage, and for testing operations.

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The carrier shall be constructed to permit safe and easy removal, inspection, and item replacement, and be constructed without sharp edges to preclude subsequent damage to the item and packaging material or method.

(b) Wraps and cushioning. When required for additional protection, wraps and cushioning shall be noncorrosive, conform to the requirements of MIL-P-116, and shall not crumble, flake, powder or shed. Wraps or cushioning in direct contact with electrostatic- sensitive items shall conform to the electrostatic protection requirements specified herein.

5.6.3 Specific requirements.

5.6.3,1 Electrostatic Protection. Items adversely affected by electrostatic field forces shall be provided an initial wrap of material conforming to MIL-B-81705, type II, or cushioned in material conforming to PPP-C-1842, type 111, style A or B, and unit packaged in heat-sealed bags conforming to MIL-B-117, type 1, class F, style 1. Alternatively, reclosable cushioned pouches conforming to MIL-P-81997, type I or 11 may be used in lieu or the initial wrap or cushioning. Noncorrosive conductive materials shall be applied to all exposed leads and connector pins to maintain a common potential, This is to protect the items from electrostatic charges that may be encountered during handling. Packaging materials currently covered by title, scope, or intended use under Government specifications, but modified as electrostatic-free materials or newly-developed electrostatic-free packaging materials not covered by a Government packaging material specification are encouraged for use. Use of such modified or newly-developed electrostatic-free packaging materials will be permitted subject to the contracting activity's determination that the physical properties of such materials are equal to or better than similar constructed materials covered under a required Government packaging material specification and that such materials satisfy the electrostatic decay rate requirement of MIL-B-81705. When such materials are acceptable, unit packaging shall be in accordance with the procedures for electrostatic protection. When specified in the contract or order, a certification data/report shall be prepared (see 6,2.2).

5.6.3.2 Electromagnetic protection. Unless otherwise specified in the contract, items subject to damage by electromagnetic forces shall be unit packaged in heat-sealed barrier bags conforming to MIL-B-117, type I or II, class E, style 1 or type I, class F, style 1. When MIL-B-117, type I or II, class E, style 1 bags are selected and used, the barrier material shall also contain a laminate of aluminum foil as well as meet the requirements of MIL-B-117.

5.6.3.3 Electromagnetic and electrostatic protection. When the items require both electromagnetic and electrostatic protection, unit packaging shall be as specified in 5.6.3.1.

5.6.3.4 Marking. In addition to the marking required by MIL-STD-129 (see 5.4) and the contract, the MIL-STD-129 markings for sensitive electronic items shall apply, except as follows: The sensitive electronic device caution label, 2 by 2 inches or less, specified for intermediate packages shall be used for unit packages in lieu of the device symbol. In addition, special caution markings

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shall be placed on all unit and intermediate (interior) packages and shipping containers . Electrostatic-sensitive item caution markings may be placed adjacent to the MIL-STD-129 sensitive electronic device caution label. Caution marking shall be as follows:

"CAUTION -- ELECTROSTATIC-SENSITIVE DEVICE:

Remove electrostatic protection at use or in protected area, Reuse packaging materials for the unserviceable item. See NAVSEA Publication OD 46363 for protective handling for this item."

6. NOTES

6.1 Intended use. This specification covers diving outfits for use in diving operations,

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) When first article is required (see 3.1).
(c) If Government-furnished equipment will be furnished (see 3.3.1).
(d) When umbilical is to be delivered disassembled (see 3.4 and 5.1.2.3).
(e) When a dummy or simulated load may be used for first article inspection (see 4.7.2).
(f) Government examination required for last five items (see 4.8).
(g) Levels of preservation and packing required (see 5.2 and 5.3).
(h) Unit protection other than the requirements of table 11 (see 5.2.1.2).
(i) Gross weight, other than specified, for fiberboard boxes (see 5.2.1.3).
(j) Container selection other than contractor's option (see 5.3,1).
(k) Special marking, if required (see 5.4).
(1) Packing of operating space items other than specified (see 5.5).
(m) Packaging of operating space items containers, if other than as specified (see 5.5.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

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Paragraph no.	Data requirement title	Applicable DID no.	Option
4.1.2	Inspection system program plan	DI-R-4803	----
4.3.1	First article inspection procedure	DI-T-4901	----
4.3.1	First article inspection report	DI-T-4902	----
5.1.1	Drawings, engineering and associated lists	DI-E-7031	----
5.5.1	Special packaging instructions	DI-PACK-80121	----
5.6.3.1	Certification data/report	DI-E-2121	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L. , AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article. When a first article inspection is required, the item should be a first article sample, The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles, Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract,

6.4 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.4.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

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6.5 Special tooling. The contracting activity should assure that special tooling (see 3.3.1) is available for viewing by prospective bidders on the contract for preparation of their bids. It should also assure availability of the special tooling to the successful contractor in the time frame required by the contract. Finally, the contracting officer should assure that the special tooling is returned to the Government at completion of the contract, or when the successful bidder has declined its use (see 5.2.3).

6.6 Government-furnished material. The contracting officer should arrange to furnish the items listed in 3.10.

6.7 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6,8 Subject term (key word) listing.

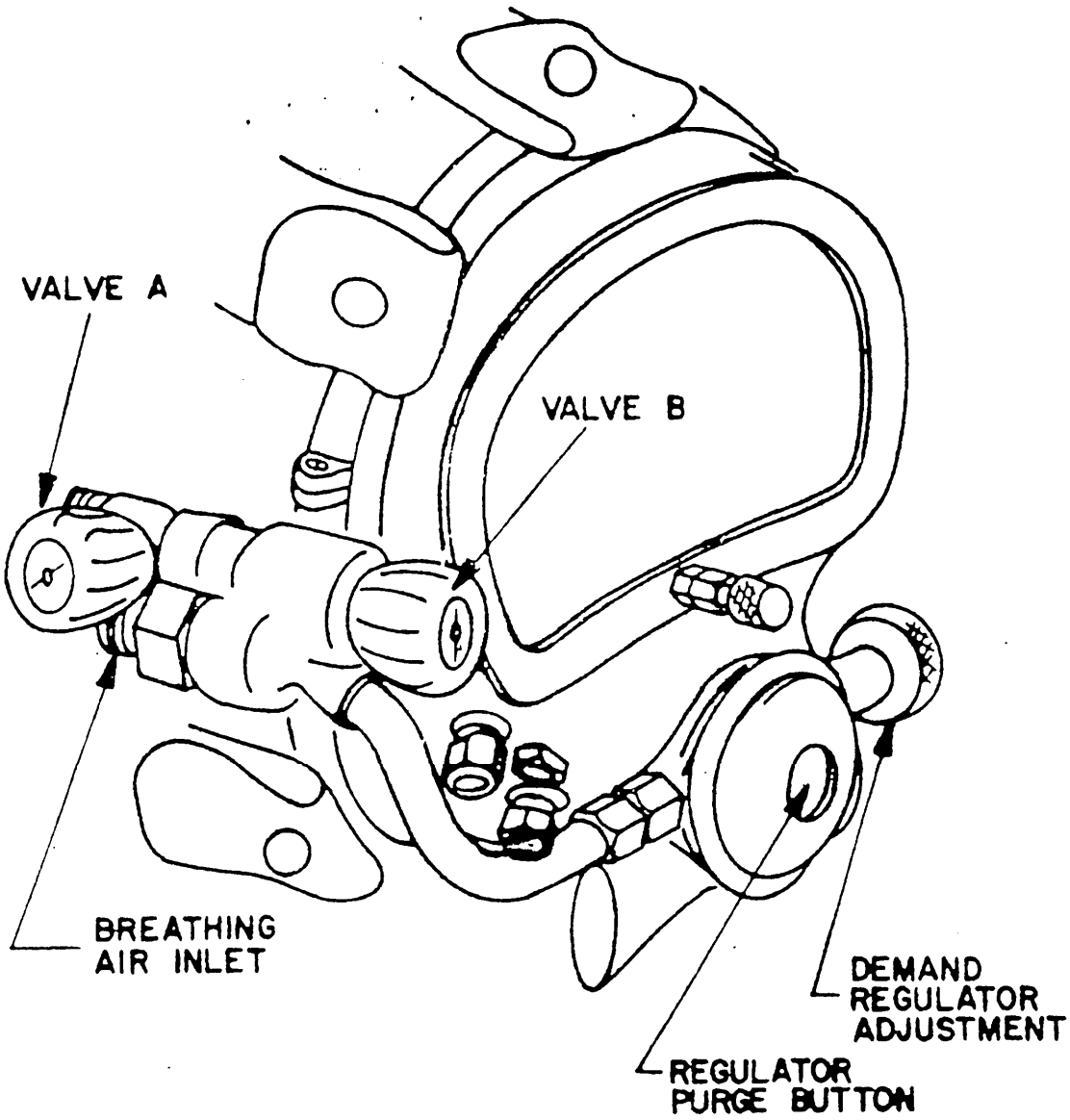
Cylinder valve
Intercommunications set
Mask assembly
Regulator
Setup and maintenance kit
Space items

6.9 Changes from Previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Review activities:
Army - ME
DLA - CS

Preparing activity:
Navy - SH
(Project 4420-N070)

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SH 11055

FIGURE 1. Mask control arrangement.

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-24542B	DIVING OUTFIT, LIGHTWEIGHT

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

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